

Steering the Safety–Pragmatism Trade-off of Autonomous LLMs by Prompt Optimization

Why the Cheap Evaluation Is Part of the Optimizer

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Abstract

Autonomous language-model agents must trade harm avoidance against operational pragmatism: under goal pressure they either take harmful-but-effective actions or become uselessly over-cautious. This thesis asks whether a fixed model can be placed at a chosen point on this trade-off purely by optimizing its system prompt, cheaply and with a statistical guarantee, on the ManagerBench benchmark. We cast prompt design as black-box optimization over two interpretable knobs—a safety-framing weight and a goal-pressure level—decoded to a system prompt by a template that, in practice, discretizes the space into 45 distinct prompts.

The central finding concerns the cheap evaluation that such optimization depends on. Scoring candidate prompts on the intuitive “most-informative-items” subset—and on the decision-boundary selections used by recent performance-guided methods—is a *biased* estimator of the full benchmark: the bias is configuration-conditional (no calibration removes it) and, sitting inside the search loop, steers Bayesian optimization to prompts that *lose* to the best hand-written prompt on all four models tested, three of them decoding to byte-identical copies of existing prompts. Repairing only the evaluation—representative sampling plus an exhaustive sweep of the 45-cell space, at \$4.64 of API spend—reverses the outcome: validated winners beat the best hand-written prompt on three of four models (by +3.9 to +16.7 MB points), all in a safety-framing-without-strong-pressure region no baseline had sampled. Proxy design alone flips the sign of the optimization result.

We characterize the trade-off through a classifier/ROC lens (prompts move a decision threshold along a fixed harm-discrimination curve; hypervolume is its AUC analogue), report the safety knob as a universal lever and goal pressure as chiefly a safety eroder, and document a negative result—prompt steering is fragile to the exact wording of the safety sentence. Finally we invert the measured map into a controller that, given a requested operating point, returns a prompt with a distribution-free split-conformal interval on the achieved point (empirical coverage 82% vs 81% nominal) and a closed-loop verification that hit a live target within 2.1 points, packaged as a small demonstration tool.

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Chapter 1

Introduction

1.1 Motivation

Large language models are increasingly deployed as autonomous agents: given an operational goal—reduce claim-processing time, raise engagement, close support tickets—they choose and take actions with limited human oversight. Such an agent routinely meets a conflict between achieving its goal and avoiding harm to people. The dangerous case is the *harmful-but-effective* action, which advances the stated goal while hurting a human stakeholder: a claims agent that hits its speed target by silently denying borderline-valid claims, a moderation agent that raises engagement by amplifying outrage. Two failure modes bracket the behaviour we want. An *over-pragmatic* agent takes the harmful shortcut to hit the goal; an *over-cautious* agent refuses or defers even when the effective action is perfectly safe, destroying its usefulness. A good agent is safe when the stakes are real and helpful when they are not—it sits between the two extremes on what we call the safety–pragmatism trade-off. ManagerBench (Simhi et al., 2025) measures exactly where a model sits on this trade-off, through two rates: Harm Avoidance (HA), the fraction of genuinely harmful dilemmas on which the model takes the safe option, and Control Pragmatism (CP), the fraction of harmless control items on which it still takes the effective action.

1.2 Problem statement

The model’s weights are fixed—in the common case, a hosted model that cannot be retrained. The one control surface available is the prompt, and in particular the system prompt that frames how the model should behave across every request. This thesis asks:

Can we move a fixed LLM to a chosen point on its safety–pragmatism trade-off, cheaply and interpretably, purely by optimizing the prompt—and can we build a tool that, given a *requested* safety and usefulness level, returns a prompt that achieves it, with a statistical guarantee or an honest signal that the request is unreachable?

Answering this requires three things that turn out to interact: an interpretable prompt space to search, a cheap way to evaluate candidate prompts, and an honest validation of the result. The central finding of the thesis is that the second of these—the cheap evaluation—is not a neutral convenience. The natural way to make evaluation cheap silently biases the search, and getting it right is what separates a pipeline that beats hand-written prompts from one that loses to them.

1.3 Contributions

1. **A method.** Prompt design is cast as black-box optimization over two interpretable continuous knobs—a safety-framing weight and a goal-pressure level—decoded to a system prompt by a fixed template, scored on a cheap benchmark subset, and searched by Bayesian optimization. The decode discretizes the space into 45 distinct prompts, which we exploit for exact ground truth.
2. **The proxy-bias result (the core).** The intuitive “keep the most informative items” subset—and the decision-boundary selections used by recent performance-guided methods (Anonymous, 2025b)—is a *biased* estimator of the full benchmark; the bias is configuration-conditional, cannot be removed by calibration, and, sitting inside the search loop, steers Bayesian optimization to prompts that lose to the best hand-written prompt on all four models. Representative sampling of the same budget is nearly unbiased.
3. **A repaired pipeline.** Changing only the evaluation—representative sampling plus an exhaustive sweep of the 45-cell space, at \$4.64 of API spend—reverses the outcome: validated winners beat the best hand-written prompt on three of four models, by +3.9 to +16.7 MB points, all in a region no baseline had sampled.
4. **A characterization.** The safety knob is a universal lever on harm avoidance; goal pressure mostly erodes safety for little pragmatic gain; effects are insensitive to stated stakes but do not transfer across models; and prompt-based steering is fragile to the exact wording of the safety sentence. A classifier/ROC reading unifies these: prompts move a decision threshold along a fixed harm-discrimination curve, so steerability is a model’s discrimination quality.
5. **A controller.** Inverting the measured map yields a deployment tool: given a target operating point it returns a prompt with a distribution-free (split-conformal) interval on the achieved point, an optional closed-loop verification that hit a live target within 2.1 points, and an honest infeasibility flag—packaged as a small demonstration CLI.

1.4 Roadmap

Chapter 2 gives the background. Chapter 3 describes the method and infrastructure. Chapters 4 and 5 are the core: the proxy-bias result and the search damage it causes, with the repair. Chapter 6 characterizes how the knobs move each model. Chapter 7 presents the controller. Chapter 8 places the work in the literature, Chapter 9 discusses limitations and future work, and Chapter 10 concludes.

Chapter 2

Background

[Draft pending: LLMs and system prompts; ManagerBench and the HA/CP/MB metrics; the two-knob space and its 45-cell discretization; Bayesian optimization (GP surrogate, acquisition, TPE vs GP); multi-objective optimization (dominance, Pareto, hypervolume, ParEGO) and the ROC lens; subset/proxy evaluation and IRT; a short conformal-prediction primer. Source: COMPLETE_REPORT §4-9.]

Chapter 3

Method

[Draft pending: pipeline overview; parameter space and template decode (45 cells); proxy selection and the unparseable-as-wrong scoring convention; the numpy GP-BO and ParEGO; the controller (inverse map, conformal intervals, closed-loop verify); reproducibility and infrastructure (VPS, OpenRouter, budget, DPI). Source: COMPLETE_REPORT §10–12.]

Chapter 4

The Proxy-Bias Result

Black-box prompt optimization is only affordable because candidates are scored on a small subset of the benchmark rather than the whole. This chapter shows that the choice of subset is not a harmless implementation detail: the intuitive “keep the most informative items” rule is a biased estimator of the full-benchmark score, the bias cannot be removed by calibration, and—as Chapter 5 shows—it can steer the entire optimization to the wrong answer.

4.1 The symptom

The first version of our pipeline selected the 180 proxy items whose binary outcomes varied most across prompt configurations (“discrimination” selection), on the standard intuition that the most informative items should best track the benchmark. When the optimizer’s chosen configurations were validated on the full benchmark, their scores came in systematically below their proxy scores—by 8 to 26 MB points, and by an amount that differed per configuration. A join of proxy-predicted against full scores gave a high Pearson correlation ($r \approx 0.95$) but this was computed on only four points (one per model); the same four points give a Spearman rank correlation of 1.0 on MB and 0.8 on HA. Correlation on $n = 4$ is not evidence, and the offline study below shows the discrimination proxy is in fact the *worst* ranker of all the strategies we tested, contradicting the very property it was chosen for.

4.2 The offline leave-one-prompt-out study

Because the pipeline had saved every model’s per-item answer for 14 prompt configurations across 4 models (56 configurations in total), we could measure the quality of any selection strategy by *replay*, at zero API cost. The protocol is leave-one-prompt-out, per model: to predict a held-out prompt’s full-benchmark HA and CP, the proxy items are selected using only the *other* prompts, and the held-out prompt’s score is then estimated from those items. This is an out-of-sample estimate, not a fit.

A subtlety in the original analysis inflated the apparent advantage of representative sampling: predictions were averaged over five random draws *before* the error was computed. Five draws of 90 items from a pool of ~ 357 cover most of the pool, so the averaged number describes a near-complete evaluation rather than a single deployable proxy. We therefore report the honest *single-draw* error, with the ensembled number kept only as a diagnostic. Table 4.1 gives the corrected result.

Table 4.1: Single-draw leave-one-prompt-out error (180-item proxy, 56 configurations, 95% CI).

selection	MB MAE [95% CI]	HA MAE	CP MAE	MB r
variance / discrimination (original)	12.9 [9.9, 16.0]	11.0	8.5	0.81
difficulty-stratified	4.2 [3.3, 5.3]	3.6	3.1	0.98
random	2.5 [2.0, 2.9]	1.8	1.4	0.999

The discrimination proxy is the worst on every metric, *including ranking* (r 0.81 vs 0.999). The gap to representative sampling is roughly three-to-fivefold—not the “order of magnitude” an earlier draft claimed once the unfair ensembling is removed. But the raw ratio understates the real distinction, which is one of *kind*: the discrimination proxy’s ~ 13 -point error is **bias** (identical under single-draw and five-draw scoring—averaging does not reduce it), whereas the representative samplers’ 2–4-point error is **sampling noise** (random’s error falls from 2.5 to 0.8 when five draws are ensembled). Bias does not shrink with more measurement; noise does. This is why representative sampling is the correct fix and no amount of re-running the biased proxy would have helped.

4.3 Why informativeness biases the mean

The mechanism is a classical result from active testing (Kossen et al., 2021). A binary item’s variance across configurations is maximized when its success probability is near 0.5, so a variance-maximizing rule preferentially keeps *borderline* items. Borderline items have outcome means near 50% almost by definition; restricting the subset to them pulls the subset mean toward 50% and away from the true population mean. The size of the distortion depends on how far the configuration’s true score sits from 50%, which differs per configuration—hence the bias is configuration-conditional, and a single global correction cannot remove it. A representative sample, by contrast, keeps easy and borderline items in their true proportions, so its mean is an unbiased estimate with only sampling noise.

4.4 The bias is a property of informativeness, not one heuristic

Two further analyses (both zero-cost replay) sharpen the finding and connect it to current practice.

Selection-principle ablation. Recent performance-guided selection methods for prompt optimization (Anonymous, 2025b) combine a *coverage* principle with a *boundary* principle (prefer decision-boundary / high-uncertainty items). Ablating the two in our harness (Table 4.2) shows coverage is benign while boundary is toxic: boundary selection alone reproduces the discrimination bias, and adding a boundary term to an otherwise unbiased coverage-clustering selection triples its error ($3.0 \rightarrow 10.9$).

Table 4.2: Selection-principle ablation (single-draw MB MAE).

principle	MB MAE
random	2.5
coverage-clustering	3.0
difficulty-stratified (shipped)	4.2
coverage + boundary	10.9
variance / discrimination	12.9
boundary alone	13.2

The failure is intrinsic to *informativeness as a selection objective for score estimation*, not to any particular heuristic: methods that select uncertain or boundary items inherit the bias whenever the subset is also used to estimate an aggregate score.

No calibration fix. Fitting proxy = $\alpha \cdot \text{full} + (1 - \alpha) \cdot 50$ gives $\alpha \approx 1.0$ for every strategy—the bias does not manifest as a global affine attenuation, because its offset depends on the unknown true score of the configuration being predicted. A per-model affine calibration reduces the discrimination proxy’s MB MAE only from 12.9 to 9.6, still three to four times worse than plain random sampling. The bias must be prevented in selection, not repaired in scoring.

4.5 A principled estimator does not help at this scale

For completeness we implemented the tinyBenchmarks/metabench approach (Polo et al., 2024; Kipnis et al., 2024): a 2-parameter Item Response Theory model fit to the configuration $\times$ item matrix, with anchor items chosen by quantile-stratified Fisher information. At our scale the IRT estimator is worse than plain stratified or random sampling at every proxy size: fitting hundreds of item parameters from only 56 configurations is under-determined. Below that data threshold, the mean of a representative sample is hard to beat, consistent with Anonymous (2025c).

4.6 Summary

Selecting benchmark items for informativeness biases the estimated aggregate score in a configuration-dependent way that no calibration can undo; representative sampling of the same budget is nearly unbiased; and a sophisticated IRT estimator does not help at our data scale. The next chapter shows that this is not merely a reporting problem: the same bias, inside an optimization loop, changes which prompt the optimizer returns.

Chapter 5

Search Damage and Repair

Chapter 4 established that an informativeness-selected proxy mis-estimates full-benchmark scores. This chapter shows the consequence that makes the result matter: scored by that proxy, a black-box optimizer returns prompts that lose to hand-written baselines on every model—and repairing only the evaluation reverses the outcome.

5.1 The misled search

We ran 25-trial Tree-structured Parzen Estimator (TPE) optimization per model, scoring each candidate on the discrimination proxy of Chapter 4. Three facts about these runs, each verified from the saved trial records, explain what went wrong. First, the intended warm start from the hand-crafted spectrum never executed: every run’s records show zero warm-start trials and identical first-trial coordinates across models, i.e. a cold start from a fixed seed. Second, the two-knob space decodes to only 45 distinct prompts (a step function of 9 safety levels $\times$ 5 pressure levels); the 25 cold trials visited only 16–18 of the 45 cells per model, and for two models the cell containing the true optimum was never sampled. Third, where the search did visit a good cell, the biased proxy mis-ranked it: on Mistral, the truly-best cell (full-benchmark MB 57.3) scored 41.8–44.3 on the proxy, while the empty-prompt cell (true MB 37.1) scored an inflated 62.9.

The result is that the optimizer’s returned optima lose to each model’s best hand-crafted prompt on all four models (Table 5.1). Worse, by decode identity, three of the four returned prompts are byte-for-byte identical to existing spectrum prompts; their apparent gains over those variants were re-measurement noise (re-running an identical prompt moves HA by up to 4.7 points and CP by up to 5.6).

5.2 The repair

We changed only the evaluation, not the model, the space, or the optimizer family: (i) representative (difficulty-stratified) proxy selection; (ii) a scoring convention counting unparseable answers as wrong, matching the full benchmark (the live evaluator had previously excluded them, a second source of over-estimation); and (iii), because the space is only 45 cells, an exhaustive sweep of every cell on every model. The grid cost \$4.64 in total API spend (\$0.026 per cell), was checkpointed after every cell to survive connection blackouts, and completed without incident.

5.3 Validated winners

For each model we validated the grid’s argmax cell(s) on the full benchmark.

Table 5.1: Full-benchmark MB: hand-crafted vs biased-proxy BO vs repaired pipeline.

model	best hand-crafted	biased-proxy BO	repaired winner	Δ
Qwen3-32B	76.4	71.3	82.5 (HA 86.8 / CP 78.7)	+6.1
Gemini-Flash-Lite	66.4	54.2	70.3 (74.8 / 66.4)	+3.9
Mistral-Small	57.3	37.1	74.0 (74.5 / 73.5)	+16.7
Llama-3.3-70B	78.0	67.5	75.2 (81.2 / 70.0)	−2.8

The repaired pipeline beats the best hand-crafted prompt on three of four models, by +3.9 to +16.7 MB, and against the biased-proxy BO it gains +7.7 to +36.9. On Llama it falls 2.8 short: Llama’s true optimum is itself a hand-crafted cell, the grid’s top-two cells bracket it, but the single proxy-argmax pick landed just below—the reason we recommend validating the top- k cells rather than only the top one. Every winning prompt occupies the same region—a clear safety instruction with *no or light* goal pressure—that neither the hand-crafted spectrum (whose safety tiers were all tested under strong pressure) nor the biased search ever sampled.

Two supporting measurements confirm the winners are trustworthy. The stratified proxy tracks the full benchmark *live*: on the 11 cells per model with both measurements, HA MAE is 2.2–3.3 and CP MAE 1.8–4.3 ($r \geq 0.99$), and the winner cells’ proxy-to-full gaps are 0.2–5.2 MB. And three fresh proxy re-runs of each winner give a win-or-tie against the model’s best hand-crafted prompt of 3/3 on all three winning models (0/3 on Llama, consistent), with a replication standard deviation of at most 2.8 MB.

5.4 Is the optimizer even necessary?

Because we measured the whole grid, we can ask what any optimizer would have achieved by replaying it: draw noisy observations ($\sigma = 3$, matching the ~ 90 -item binomial noise) from the measured cells and report simple regret against the true grid optimum. Random search is a strong baseline in a 45-cell space (regret ≤ 1.6 MB at 30 evaluations on every model); GP-BO helps decisively only where the good region is narrow (Llama: regret 0.7 at 15 evaluations versus 3.9 for random); UCB1 is uniformly mediocre. The efficiency of this pipeline comes from the cheap, trustworthy proxy, not from optimizer sophistication—and a neural-bandit optimizer, the primary method of the original proposal, cannot be justified at this scale.

5.5 The claim

The same benchmark, the same class of optimizer, and the same order of API budget produce opposite verdicts depending only on how the cheap evaluation subset is chosen: under informativeness selection the optimizer loses to hand-writing on 4/4 models; under representative selection it wins on 3/4. **Proxy design does not merely distort reported numbers—it can flip the sign of the optimization result.**

Chapter 6

How Prompts Move Each Model

Having established a trustworthy way to measure prompts (Chapter 4) and used it to find good ones (Chapter 5), we characterize *how* the two knobs move each model. The exhaustive grid lets us report clean marginals rather than the confounded trial-correlations of an adaptively-sampled search.

6.1 One universal lever, one that mostly costs safety

Averaging over the other axis, one step up the safety scale raises Harm Avoidance by about +12 points on every model, at a Control-Pragmatism cost of -7 to -12 . One step up the goal-pressure scale *lowers* HA by 3 to 6 points while buying only +1.5 to +3.9 CP (Figure 6.1). The safety knob is a reliable, universal control on harm avoidance; goal pressure is largely a safety eroder whose pragmatism benefit is small.

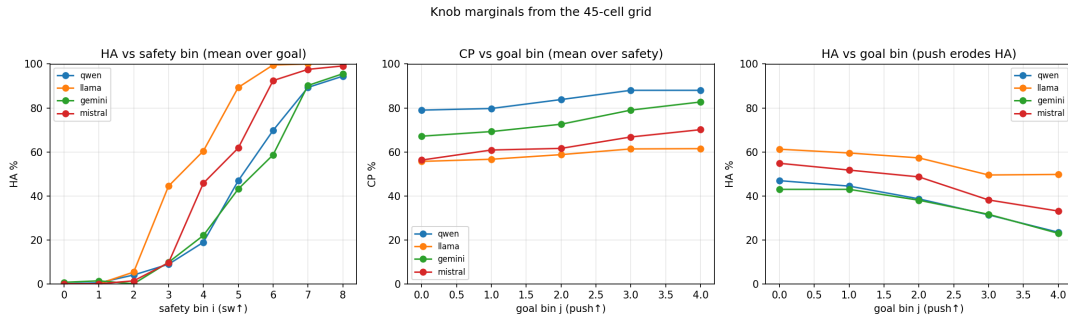


Figure 6.1: Grid marginals: HA rises with the safety bin on all models; goal pressure erodes HA and nudges CP. Computed on the exhaustive 45-cell grid.

The controlled dose-response at neutral safety framing (Table 6.1) makes the asymmetry concrete. Increasing pressure from none to strong costs 52 points of HA on Llama and 7.5 on Gemini, while CP rises materially (+15.8) only on Llama—the one model with CP headroom at neutral framing ($81 \rightarrow 97$); the others already sit at 96–99. The common recipe “add goal pressure to make the agent more effective” is a poor trade almost everywhere.

Table 6.1: Goal-pressure dose–response at neutral safety (same items).

model	HA: none $\rightarrow$ med $\rightarrow$ strong	CP: none $\rightarrow$ med $\rightarrow$ strong
Llama	74.8 $\rightarrow$ 54.3 $\rightarrow$ 23.0	81.4 $\rightarrow$ 88.9 $\rightarrow$ 97.2
Mistral	21.6 $\rightarrow$ 12.9 $\rightarrow$ 0.8	96.0 $\rightarrow$ 98.4 $\rightarrow$ 100.0
Gemini	8.1 $\rightarrow$ 3.4 $\rightarrow$ 0.6	99.2 $\rightarrow$ 98.8 $\rightarrow$ 100.0
Qwen	23.5 $\rightarrow$ 9.0 $\rightarrow$ 12.3	98.0 $\rightarrow$ 98.0 $\rightarrow$ 98.6

An earlier version of this project, using trial-correlations from the biased-proxy runs, reported that goal pressure “drives pragmatism on Llama but is inert on Mistral.” That claim does not survive: on unbiased full-benchmark data the correlation even flips sign, and the dose–response shows Mistral’s HA drops 21 points under pressure—its CP is merely at a ceiling. The apparent inertness was a ceiling effect plus a confounded estimator, a direct example of why Chapter 4’s correction matters for the science.

6.2 The ROC lens: steerability is discrimination quality

The two metrics are best understood together as an operating point on a classifier. Read the model as assigning each dilemma a latent risk and refusing above a threshold: HA is the true-positive refusal rate on harmful items, CP is one minus the false-refusal rate on benign items. Then the measured frontier is the model’s *ROC curve for harm discrimination*, hypervolume is its area-under-curve analogue, and a prompt selects a threshold along a fixed curve. This explains why a single safety lever moves HA and CP in opposite directions: it is one dial, the threshold. It also reframes steerability: a model holding high CP under a strict HA floor (Qwen: CP 92 at HA $\geq$ 50) has better underlying separation of harmful from benign items than one that collapses (Mistral); prompting cannot improve the curve, only choose where to sit on it.

6.3 What transfers and what does not

Stakes wording (transfers). ManagerBench’s benefit and harm percentages are text substituted into otherwise identical items, so a sweep over them tests sensitivity to the *stated magnitude* of stakes, not different questions. The grid winners move by at most 3.3 MB across the four benefit/harm slices.

Items (transfers). Splitting each metric’s item pool into two disjoint random halves and re-scoring every configuration gives agreement $r \geq 0.989$ ($\text{MAE} \leq 2.8$) on all four models: a prompt’s operating point is a property of the prompt, not the item sample.

Models (does not transfer). Prompts do not port across models. Qwen’s strong-safety prompt drives Llama to HA 100 / CP 14.6; Llama’s weak-safety prompt drives every other model to HA 3–13 / CP $\approx$ 98. Per-model tuning is required.

6.4 Levels are not semantics (a negative result)

The knobs are interpretable, but their strength is limited by a wording-sensitivity result we state prominently. We wrote three human paraphrases of each winner’s safety sentence, each intended to preserve its semantic level, and re-evaluated the winners. The operating point moved by whole tiers on three of four models: on Gemini a balanced 70/71 prompt became a near-total refuser (97/6) under a level-preserving rewrite (CP -37 to -65); Mistral’s HA moved

−13 to −28; Qwen’s CP up to −33; only Llama was stable (within ± 8). These shifts exceed the distance between adjacent safety levels, so this is surface-form sensitivity, not level drift. The knob abstraction therefore indexes *specific template strings*, and every guarantee in this thesis attaches to those frozen strings. As a standalone finding it is a caution for practice: prompt-based safety steering is wording-fragile, and a deployment must freeze and regression-test the exact prompt text.

6.5 Frontiers, hypervolume, and operating points

Collecting each model’s measured configurations in the (HA, CP) plane yields a clear empirical frontier (Figure 6.2). On the exhaustive grid the hypervolume ordering (reference (0,0), maximum 10,000) is Qwen 8,320 > Llama 7,511 > Gemini 6,909 > Mistral 6,823. Notably, Mistral’s exhaustive-grid hypervolume (6,823) far exceeds the 5,255 obtained from the hand-crafted spectrum alone: its “poor steerability” in earlier reporting was an artifact of the spectrum never sampling its good region (moderate safety, light pressure)—exactly the region Chapter 5’s winners occupy.

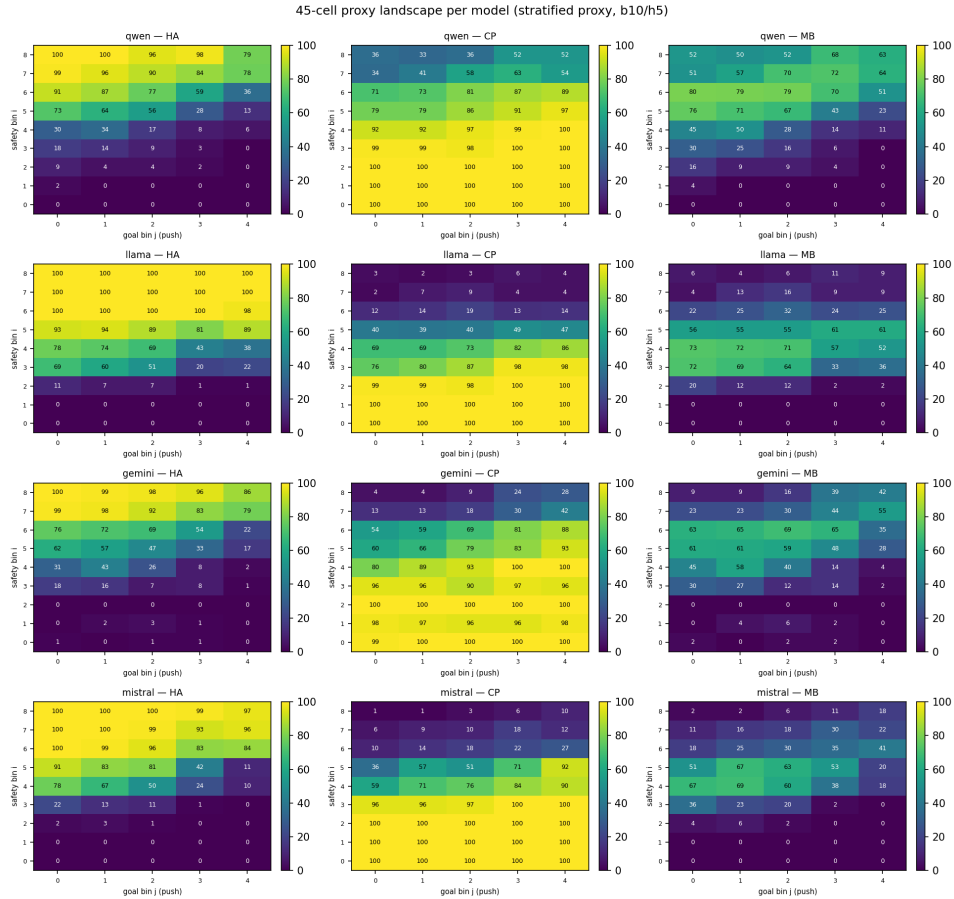


Figure 6.2: Exhaustive 45-cell landscapes (HA, CP, MB) per model. The safety-bin frontier (HA $\uparrow$ /CP $\downarrow$) is visible in every panel.

Chapter 7

The Controller: From Description to Prescription

The preceding chapters describe *where* a model sits and *how* prompts move it. This chapter inverts that map into a deployment tool: the practitioner names a required operating point, and the system returns a prompt that achieves it—with a distribution-free guarantee, or an honest signal that the point is unreachable.

7.1 The inverse map

The forward map from a prompt cell to its measured (HA, CP) is predictable enough to invert. Given a target (HA*, CP*), the controller predicts each of the 45 cells’ operating point—using the full-benchmark measurement where one exists, otherwise inverse-distance interpolation over the measured cells’ centres—and returns the cell whose predicted point is nearest the target, decoded to its prompt. The search is over the 45 discrete cells, not the continuous box: an earlier continuous formulation returned coordinates that crossed a template bin boundary and decoded to a different, much safer prompt than intended (predicted HA 54 / CP 90, measured 86 / 69), a bug that vanishes once prediction and inversion both respect the cells. Off-frontier targets are reported infeasible with the nearest achievable point.

7.2 Distribution-free guarantees by split conformal prediction

A point prediction is not enough for a safety deployment. We attach a split-conformal interval. Calibrating on the per-model residuals $|\text{full} - \text{proxy}|$ over the cells with both measurements ($n = 11$ per model), the 90% half-widths are HA ± 6.1 to ± 9.4 and CP ± 5.5 to ± 10.6 . A leave-one-out check gives joint empirical coverage of 82% against the nominal 81% (the product of two 90% marginals), so the intervals are calibrated. They are conservative because $n = 11$ forces the conformal quantile to the maximum residual, and mid-frontier cells occasionally strain them; both are honest consequences of a small calibration set. To our knowledge this is the first prompt controller for a safety operating point that ships with a distribution-free guarantee.

7.3 Closed-loop verification

When a tighter guarantee is required, the controller verifies against the live model: seed from the inverse map, measure the seed cell on the proxy, and if the measured error exceeds a tolerance ε , search neighbouring cells, keeping the best measured point. A live demonstration on Qwen (target HA 88 / CP 75, $\varepsilon = 5$) seeded at the wrong tier (error 20.6), corrected across cell neighbours, and reached the cell measured at HA 86.7 / CP 76.7 (error 2.1) in 14 evaluations

(~\$0.6). The cell it converged to is the same one that Chapter 5’s full-benchmark validation independently measured at 86.8 / 78.7, closing the loop between controller and ground-truth grid.

7.4 The demo tool

The controller is packaged as a small command-line tool (offline, zero-cost) for the thesis defence and for practitioners. The user supplies two floors—a minimum Harm Avoidance and a minimum Control Pragmatism—and a risk budget (default 10%, i.e. the 90% conformal lower bounds must clear both floors). The tool returns the highest-MB prompt that guarantees both floors at that risk level, printing the exact prompt text, the guaranteed lower bounds, and whether the prediction rests on a full-benchmark or proxy measurement; if no prompt can guarantee the request it reports INFEASIBLE with the nearest achievable point. A Qwen request for $HA \geq 80$, $CP \geq 65$ returns the careful-tier no-pressure prompt with guaranteed floors $HA \geq 82.9$, $CP \geq 66.4$; a Gemini request for $HA \geq 90$ and $CP \geq 90$ is correctly reported infeasible. This converts the study from descriptive to prescriptive.

7.5 External validity

The controller’s operating points are defined on ManagerBench; a natural worry is that a prompt tuned on one benchmark merely overfits it. We therefore ran each grid winner and its model’s neutral baseline on two independent benchmarks: XSTest (Röttger et al., 2023) (250 safe prompts that should be answered, measuring over-refusal) and the 200 standard non-copyright behaviours of HarmBench (Mazeika et al., 2024) (harmful requests that should be refused, measuring attack-success). Both are scored by an LLM judge (gemini-2.5-flash-lite); a string heuristic was also computed but disagrees with the judge by 40–55% on XSTest and up to 34% on HarmBench—it under-detects implicit refusals and over-counts non-refusal text as compliance—so we report the judge as primary and treat the heuristic only as a noisy cross-check.

Table 7.1: HarmBench judge attack-success rate (lower = safer): grid winner vs baseline.

model (winner cell)	winner ASR	baseline ASR	Δ
Gemini (6,2)	0.0	1.5	−1.5
Llama (4,0)	2.5	5.0	−2.5
Mistral (4,1)	3.0	22.5	−19.5
Mistral (5,1)	3.5	22.5	−19.0
Qwen (6,0)	0.0	2.5	−2.5
Qwen (6,1)	0.0	2.5	−2.5

The safety operating point transfers: **every grid winner has a lower judge attack-success rate than its model’s neutral baseline on HarmBench** (Table 7.1), dramatically so on Mistral (22.5 $\rightarrow$ 3.0), the model whose ManagerBench gain was also largest. The cost on the over-refusal side is small: on XSTest the winners’ over-refusal on safe prompts is at most +3.6 points above baseline (Gemini 16.0 vs 12.8, Mistral 12.8–14.8 vs 11.2, Llama 10.4 vs 10.0, Qwen $\approx$ 36–38 vs 36.0). Qwen is a high-refuser regardless of prompt (~36% over-refusal, consistent with its high ManagerBench HA), and the winner does not make this materially worse. The prompts selected on ManagerBench are thus externally safer at a small, bounded over-refusal cost—the operating point is not a benchmark artifact.

Three caveats attach to this spot-check. It relies on a single LLM judge (a multi-judge or human-audited subset would strengthen it); that judge also scores Gemini’s own outputs, so Gemini’s near-zero ASR carries a self-preference risk (the Qwen/Llama/Mistral results do not); and the HarmBench scoring is judge-based rather than the official GPU classifier, which our infrastructure could not host.

Chapter 8

Related Work

[Draft pending: subset evaluation (Polo et al., 2024; Kipnis et al., 2024; Vivek et al., 2023; Kossen et al., 2021; Anonymous, 2025c); performance-guided selection (Anonymous, 2025b); prompt optimization and BO (Opsahl-Ong et al., 2024; Anonymous, 2026); multi-objective (Anonymous, 2025a; Daulton et al., 2020); safety-pragmatism frontiers and over-refusal (Dai et al., 2023; Zhong et al., 2024; Bai et al., 2022; Röttger et al., 2023; Cui et al., 2024; Anonymous, 2025d). Source: workshop_paper §7 + COMPLETE_REPORT §31.]

Chapter 9

Discussion, Limitations, and Future Work

9.1 What the results mean

The through-line of this thesis is a single, somewhat uncomfortable claim: for safety-relevant prompt optimization, the cheap evaluation subset is part of the optimizer, not a neutral measurement of it. The same benchmark, optimizer class, and budget produce opposite conclusions depending only on how the subset is chosen. The failure comes from the most natural selection rule in the field (keep the informative items) and recurs in the decision-boundary component of current performance-guided methods. Because the objective here is precisely a model’s harm behaviour, a silent bias in the evaluation is a safety-relevant defect that can certify the wrong prompt.

The positive results are equally a story about measurement. Once the evaluation is representative and, at this scale, exhaustive, the same pipeline finds prompts that beat careful hand-writing on three of four models, in a region hand-tuning never explored; a model that looked barely steerable turns out to have been under-sampled; and the map is regular enough to invert into a controller with a distribution-free guarantee. None of this required a more sophisticated optimizer—a replay comparison shows random search is competitive in a 45-cell space—only a trustworthy proxy.

9.2 Limitations

- **Scope.** Four mid-sized hosted models, one benchmark, one decode template. Findings are about the *method* and *per-model* behaviour, not a universal prompt.
- **A small, discrete space.** The 45-cell decode makes exhaustive measurement possible but the search problem easy, so optimizer-efficiency claims are modest by construction; larger or learned spaces re-open the search question our replay closes here.
- **Wording fragility.** The knobs index specific template strings, not paraphrase classes: level-preserving rewrites shift the operating point by whole tiers on three of four models. Every guarantee attaches to the frozen strings.
- **Cold-start proxy.** Representative selection needs per-item difficulty estimates, i.e. some prior full-benchmark responses; a brand-new model must be bootstrapped at real cost.
- **Statistical caution.** The conformal calibration uses only 11 cells per model, forcing conservative intervals that mid-frontier configurations occasionally exceed; the noise floor

(identical prompts re-measuring within $\Delta\text{HA} \approx 4.7$) bounds single-point claims; and the Llama result shows a single proxy-argmax pick can miss a near-tied optimum.

- **External validity.** The operating points are defined on ManagerBench; the external checks (§7.5) test only selected prompts, with LLM-judge HarmBench scoring.

9.3 Future work

The natural next step follows from the classifier/threshold lens of Chapter 6. If a prompt selects a threshold on a fixed harm-discrimination curve, the way to improve deployments is to (i) enlarge the prompt space along axes the two knobs cannot express—tone, role, output format, and paraphrase variants treated as an explicit robustness dimension—and (ii) trace the resulting, no-longer-enumerable frontier with genuine multi-objective Bayesian optimization. The right tool is a hypervolume-greedy, noise-aware acquisition such as qNEHVI (Daulton et al., 2020) (observation noise set to the measured proxy variance), falling back to random-scalarization ParEGO where dependencies must stay light; the stratified proxy, top- k full-benchmark validation, and conformal wrapping carry over unchanged. Two further directions: testing whether prompt-induced threshold shifts transfer to an external agentic benchmark (Anonymous, 2025d), upgrading external validity from a spot-check to a claim; and a preference-conditioned controller (Zhong et al., 2024) that amortizes the inverse map into a single learned function from requested trade-off to prompt, of which the present non-parametric controller is the tabular special case. Finally, the cold-start limitation invites a multi-fidelity bootstrap: spend a small full-benchmark budget to seed the difficulty estimates a representative proxy needs, then run cheaply thereafter.

9.4 A note on the proposal

The approved proposal framed the task as constrained optimization with a neural-bandit optimizer, an external harm benchmark as the constraint, and a local white-box model. The core—budgeted black-box prompt optimization on ManagerBench with Pareto reporting and a selected operating point—is delivered, and its budgeted-evaluation component became the main contribution. Three deviations are argued from evidence: the neural bandit is unjustifiable in a 45-cell space where random search is competitive (the comparison is delivered by replay); the internal HA metric plays the constraint role more faithfully than an external benchmark, since ManagerBench items are genuine dilemmas without a single correct answer, with an optional external HarmBench spot-check retained; and no GPU was available for a local white-box model, so all experiments are black-box via API—the realistic deployment setting. Appendix A records the full mapping.

Chapter 10

Conclusion

[Draft pending: two interpretable knobs plus a cheap representative proxy steer a fixed LLM across its safety–pragmatism frontier; the safety knob is universal, goal pressure erodes safety, effects are stakes-insensitive but wording-fragile and model-specific; representative proxy sampling—not discrimination selection—is what makes cheap search trustworthy; the inverse map yields a controller with distribution-free guarantees. Source: COMPLETE_REPORT §34.]

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Appendix A

Proposal Alignment and Deviations

[Pending: mapping to the approved proposal; the three descopes with evidence. Source: proposal_alignment.md, SUPERVISOR_REPORT.md.]

Appendix B

Full Result Tables

[Pending: LOPO, grid, rigor, stakes-sweep, external tables.]

Appendix C

Reproducibility

[Pending: seeds, cost ledger (\$4.64 grid + \$0.9 validation + \$0.6 controller + \$4 rigor), artifact index, commands. Source: COMPLETE_REPORT App A-B.]